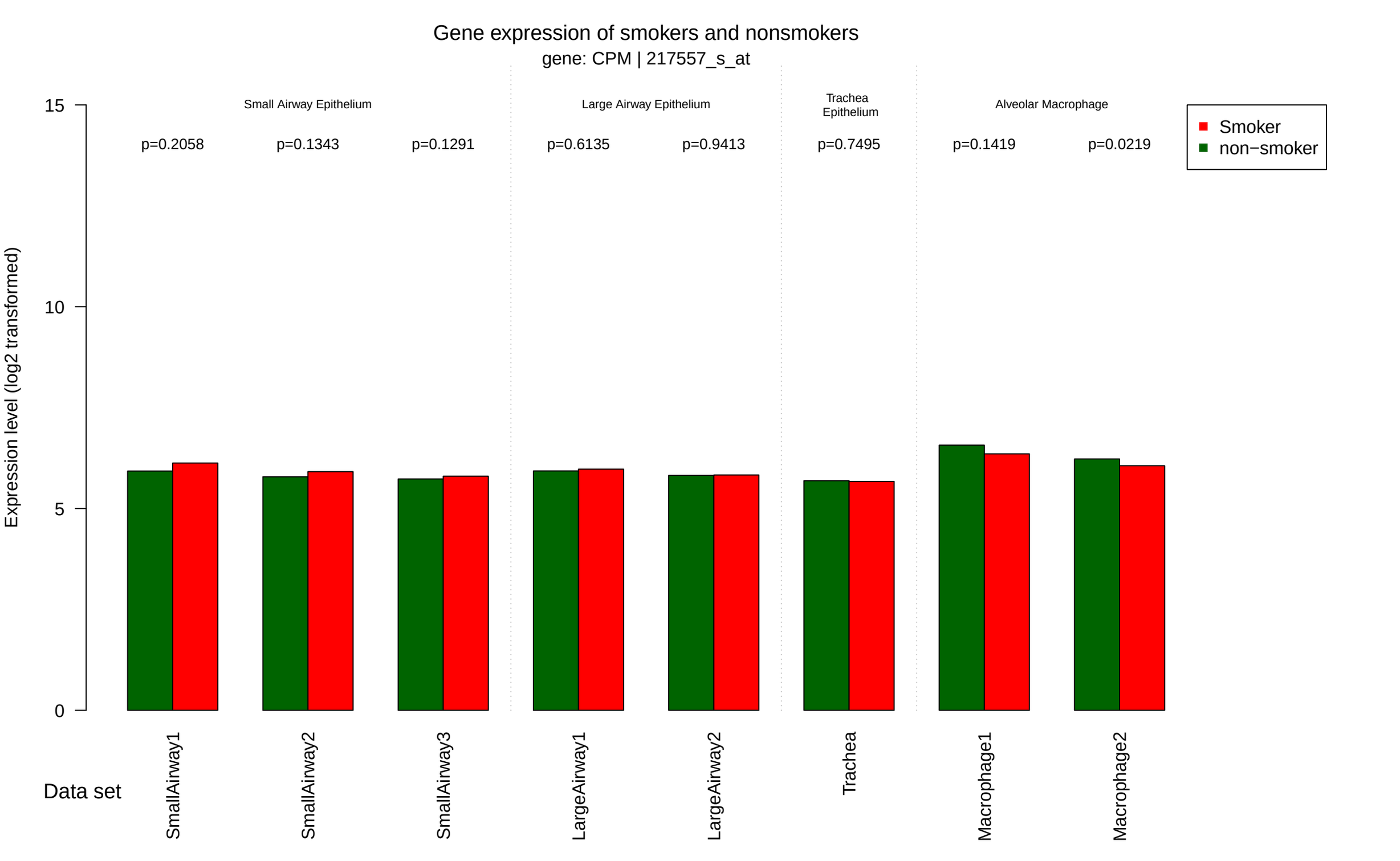
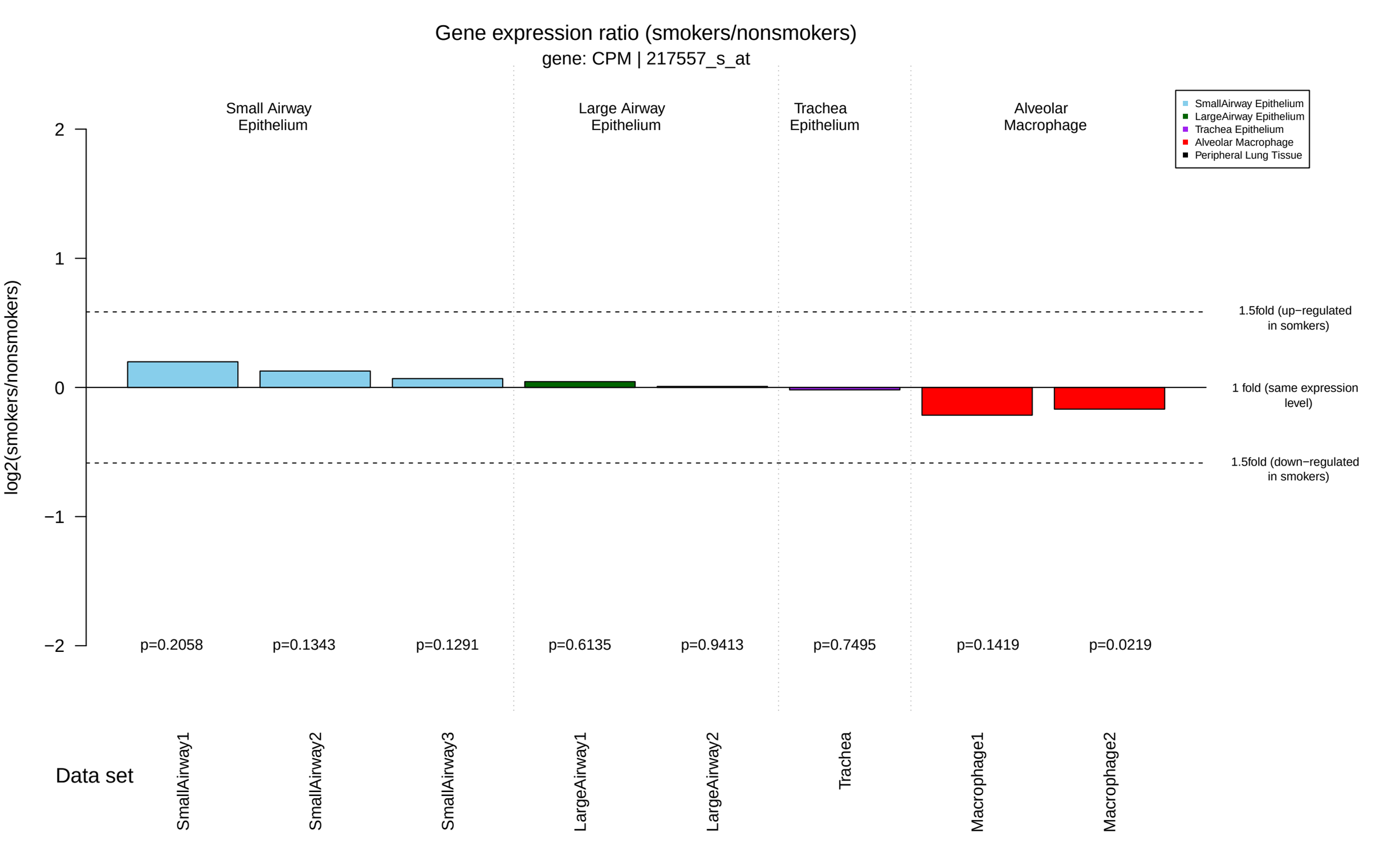
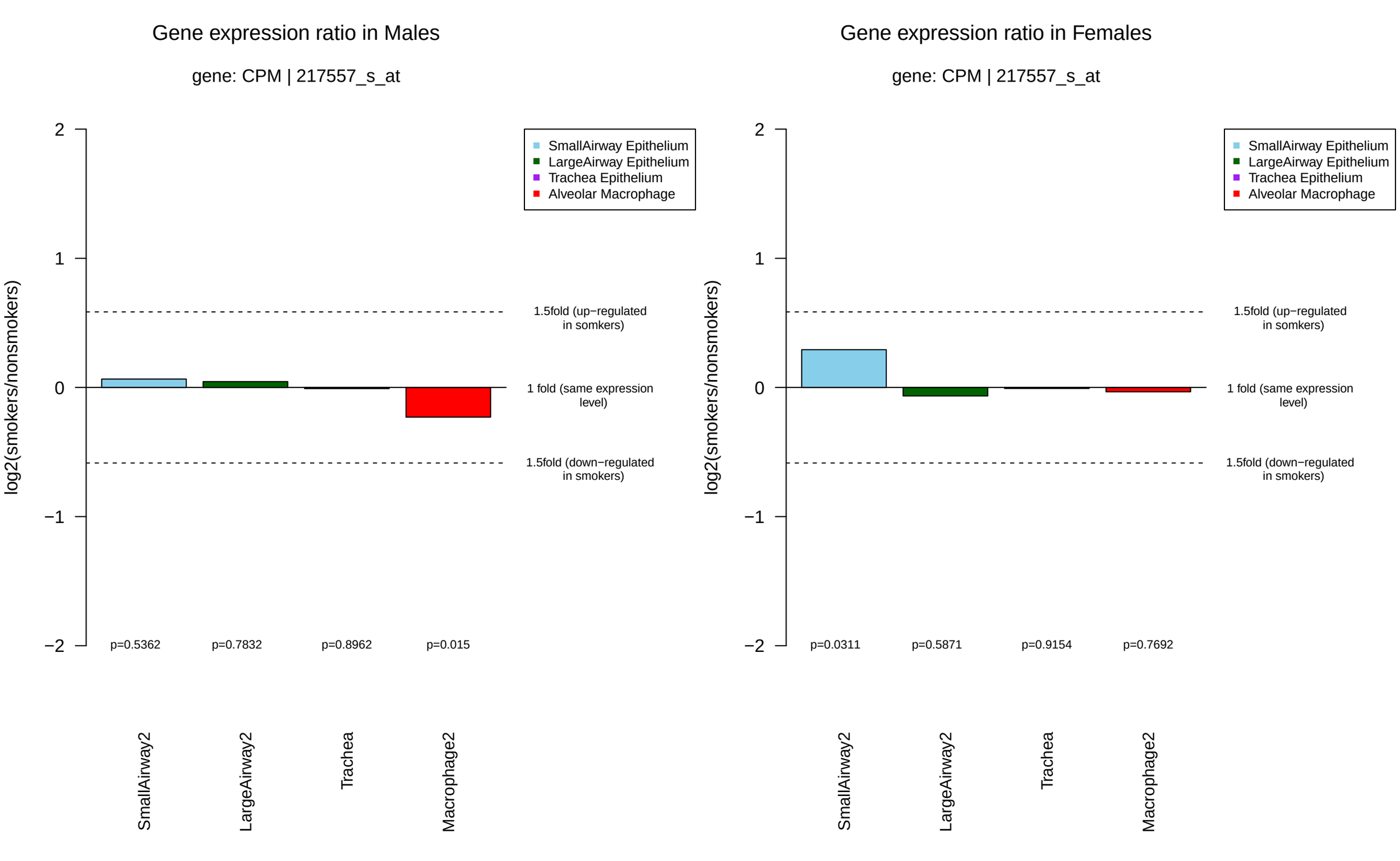
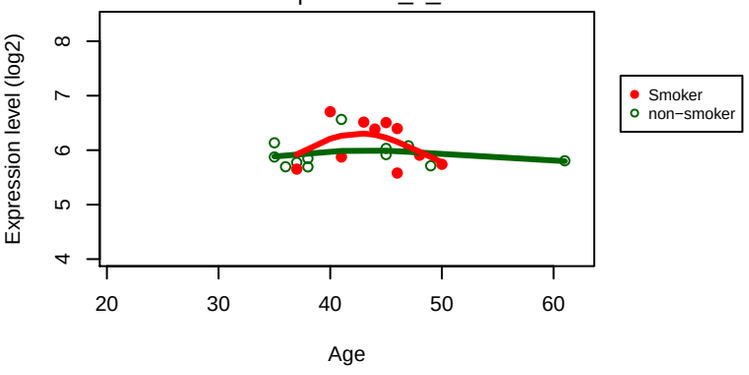
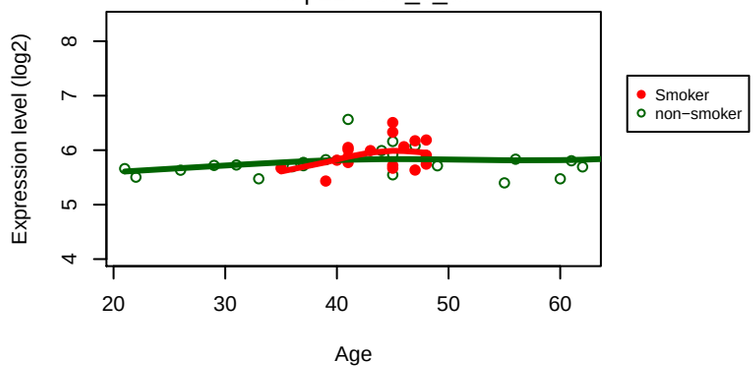
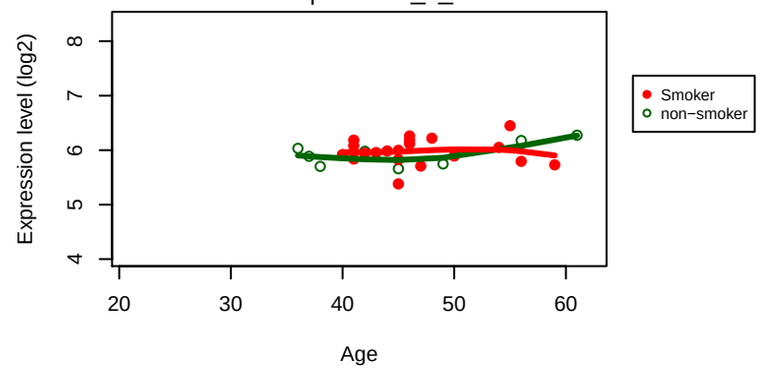

Gene expression of smokers and nonsmokers
gene: CPM | 217557_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2058
p=0.1343
p=0.1291
p=0.6135
p=0.9413
p=0.7495
p=0.1419
p=0.0219
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

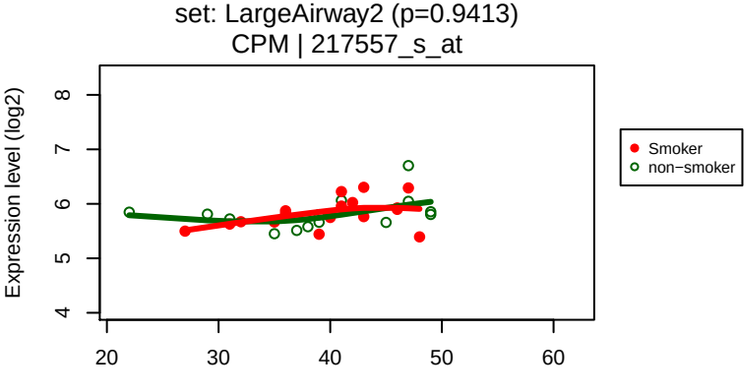

Gene expression ratio (smokers/nonsmokers)
gene: CPM | 217557_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2058
p=0.1343
p=0.1291
p=0.6135
p=0.9413
p=0.7495
p=0.1419
p=0.0219
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

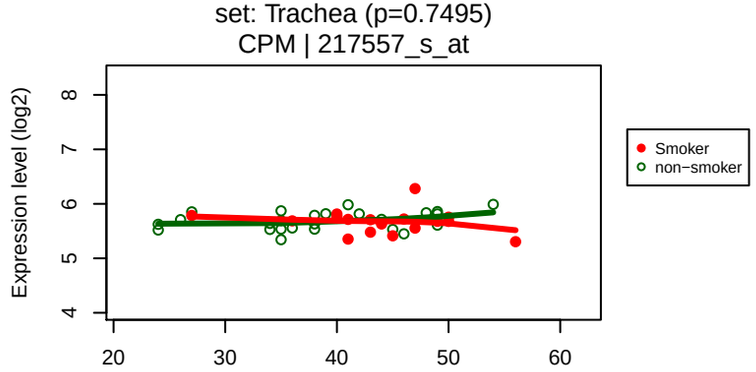

Gene expression ratio in Males
gene: CPM | 217557_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5362
p=0.7832
p=0.8962
p=0.015
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CPM | 217557_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0311
p=0.5871
p=0.9154
p=0.7692
SmallAirway2
LargeAirway2
Trachea
Macrophage2

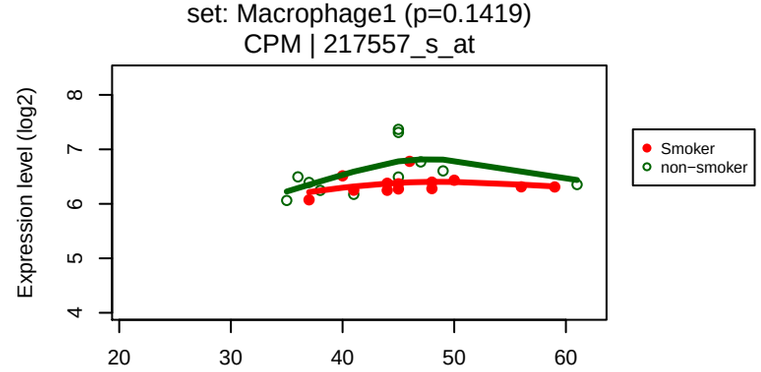

set: SmallAirway1 (p=0.2058)
CPM | 217557_s_at
Expression level (log2)
Age
Smoker
non−smoker

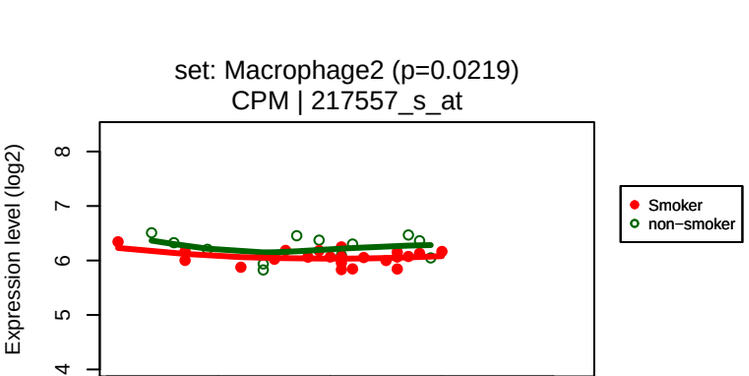

set: SmallAirway2 (p=0.1343)
CPM | 217557_s_at
Expression level (log2)
Age
Smoker
non−smoker

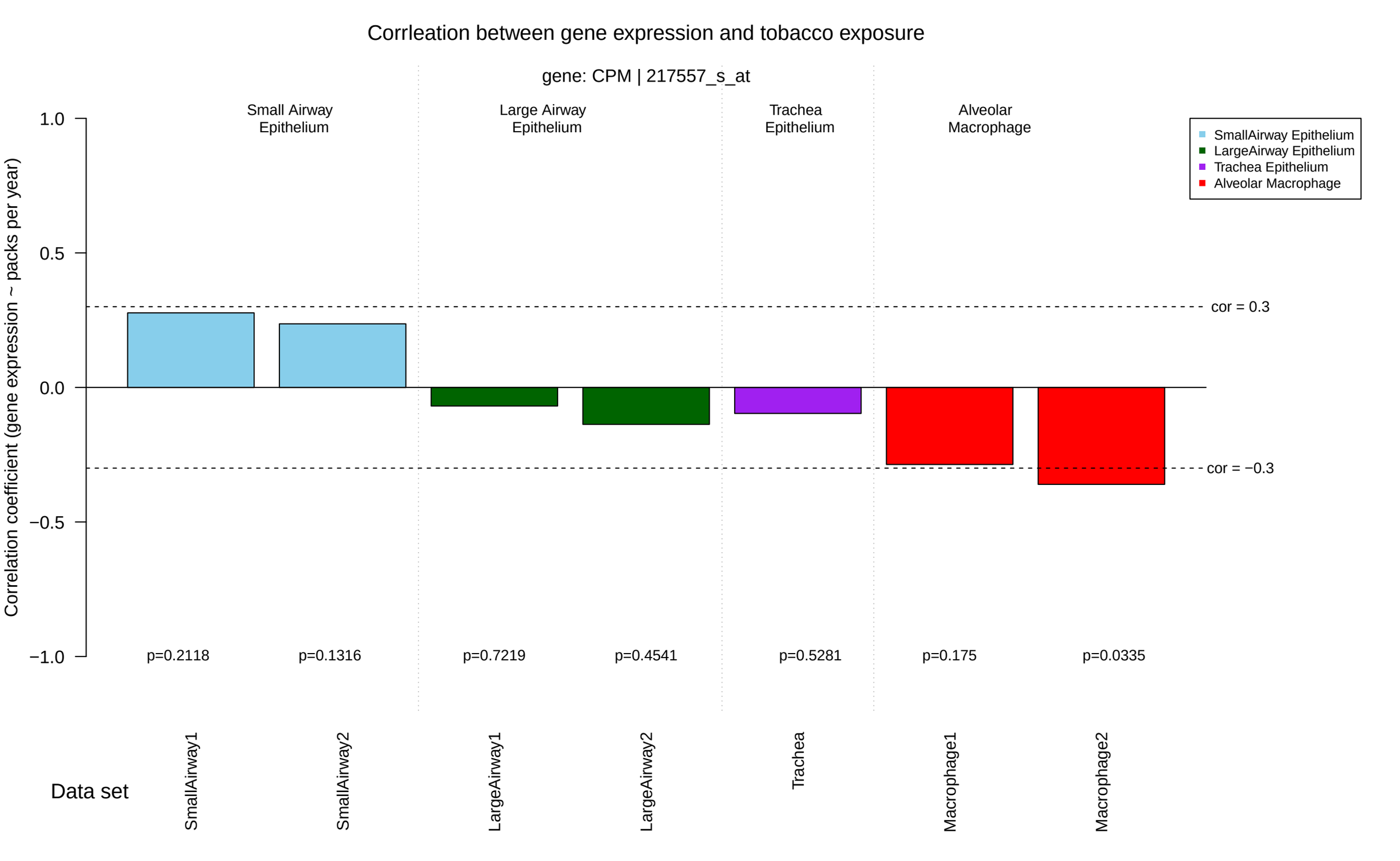

set: LargeAirway1 (p=0.6135)
CPM | 217557_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9413)
CPM | 217557_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7495)
CPM | 217557_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1419)
CPM | 217557_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0219)
CPM | 217557_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CPM | 217557_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.2118
p=0.1316
p=0.7219
p=0.4541
p=0.5281
p=0.175
p=0.0335
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2